

2024-10-03-DualSimplexMatrix

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December 31, 2024

Slide 1

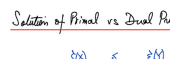
Content

Solution of Primal vs Dual Problems

Description

Slide 1 of 15 in a Linear Programming course introduces the concept of primal and dual problems. The slide uses a number line to visually represent the relationship between the optimal objective function values of the primal and dual problems. $\xi(x)$ represents the objective function value of a feasible solution x to the primal problem, and $\xi(y)$ represents the objective function value of a feasible solution y to the dual problem. The number line shows that $\xi(x) \leq \xi^*$ and $\xi(y) \geq \xi^*$, where ξ^* is the optimal objective function value for both the primal and dual problems (strong duality). The blue arrows indicate that the primal objective function values are less than or equal to the optimal value, and the dual objective function values are greater than or equal to the optimal value. The red arrows show the convergence of the primal and dual objective function values to the optimal value ξ^* . The title "Solution of Primal vs Dual Problems" clearly indicates the slide's topic. The context of the course is crucial for understanding the relationship between primal and dual problems and the concept of strong duality.

Figures



(a) A number line showing the relationship between primal and dual solutions.

Slide 2

Content

Dual Simplex Method [V] p.71

We begin with an example:

Step 0

(P)

maximize $-x_1 - x_2$ $\leftarrow$ all neg. coeffs.

subject to $-2x_1 - x_2 \leq 4$

$-2x_1 + 4x_2 \leq -8$ $\leftarrow$ original not P-feasible

$-x_1 + 3x_2 \leq -7$

$x_1, x_2 \geq 0$

The dual of this problem is

(D)

minimize $4y_1 - 8y_2 - 7y_3$

subject to $-2y_1 - 2y_2 - y_3 \geq -1$

$-y_1 + 4y_2 + 3y_3 \geq -1$

$-y_1 + y_2 + y_3 \geq 0$

$y_1, y_2, y_3 \geq 0$

Introducing variables $w_i, i = 1, 2, 3$, for the primal slacks and $z_j, j = 1, 2$, for the dual slacks, we can write down the initial primal and dual dictionaries:

Description

Slide 2 of 15 in a Linear Programming course introduces the dual simplex method using an example. The slide begins by presenting a primal linear programming problem (P), which is a maximization problem with three constraints. Crucially, the slide highlights that the original problem is not primal-feasible because the right-hand side of two constraints are negative. The objective function coefficients are all negative, which is a condition often required to start the dual simplex method. The dual problem (D) is then formulated, which is a minimization problem. The slide notes that the dual problem is constructed by introducing dual variables (y_i) and transforming the primal constraints into dual constraints. The slide concludes by mentioning the introduction of slack variables (w_i for the primal and z_j for the dual) to prepare for the simplex method iterations. The circled (P) and (D) clearly label the primal and dual problems. The red arrows and annotations explain the infeasibility of the primal problem and the reason for using the dual simplex method. The context of the course is essential for understanding the relationship between primal and dual problems and the application of the dual simplex method to solve infeasible primal problems.

Figures

Dual Simplex Method [V] p

We begin with an example:

Step 0

(P)

maximize $-x_1 - x_2$ $\leftarrow$

subject to $-2x_1 - x_2 \leq 4$ $\leftarrow$

$-2x_1 + 4x_2 \leq -8$ $\leftarrow$

$-x_1 + 3x_2 \leq -7$ $\leftarrow$

$x_1, x_2 \geq 0$

The dual of this problem is

(D)

minimize $4y_1 - 8y_2 - 7y_3$

subject to $-2y_1 - 2y_2 - y_3 \geq -1$

$-y_1 + 4y_2 + 3y_3 \geq -1$

$-y_1 + y_2 + y_3 \geq 0$

$y_1, y_2, y_3 \geq 0$

Introducing variables $w_i, i = 1, 2, 3$, for the primal slacks and dual slacks, we can write down the initial primal and dual dictionaries:

(a) Dual Simplex Method example.

Slide 3

Content

Dual Simplex Method [V] p.71

Step 0

(P)

$$\zeta = -1x_1 - 1x_2$$

$$w_1 = 4 + 2x_1 + x_2$$

$$w_2 = -8 + 2x_1 - 4x_2$$

$$w_3 = -7 + x_1 - 3x_2$$

(D)

$$-\zeta = -4y_1 + 8y_2 + 7y_3$$

$$z_1 = 1 - 2y_1 - 2y_2 - y_3$$

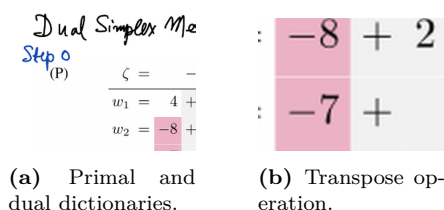
$$z_2 = 1 - y_1 + 4y_2 + 3y_3$$

origin is D-feasible

Description

Slide 3 of 15 in a Linear Programming course demonstrates the initial step (Step 0) of the dual simplex method. It presents a primal linear program (P) in dictionary form, where the objective function is $\zeta = -x_1 - x_2$ and the constraints are expressed as w_i equations. Note that w_2 and w_3 have negative right-hand sides, indicating that the primal problem is infeasible. The corresponding dual linear program (D) is also shown in dictionary form, with the objective function $-\zeta = -4y_1 + 8y_2 + 7y_3$ and constraints expressed as z_i equations. The slide highlights that the origin ($y_1 = y_2 = y_3 = 0$) is dual-feasible because all $z_i \geq 0$. The arrow indicates the transformation from the primal to the dual problem, which involves taking the transpose of the coefficient matrix and changing the direction of the inequalities. The pink shading highlights the negative right-hand sides in the primal dictionary and the positive coefficients in the dual objective function. The notation [V] p.71 likely refers to a specific page in a textbook or reference material. The context of the course is crucial for understanding the dual simplex method and its application to solving linear programs that are infeasible in their primal form.

Figures



Slide 4

Content

Dual Simplex Method [V] p.71

Step 1

(P)

$$\zeta = -4 - 0.5w_2 - 3x_2$$

$$w_1 = 12 + w_2 + 5x_2$$

$$x_1 = 4 + 0.5w_2 + 2x_2$$

$$w_3 = -3 + 0.5w_2 - x_2$$

(D)

$$-\zeta = 4 - 12y_1 - 4y_2 + 3y_3$$

$$y_2 = 0.5 - y_1 - 0.5z_1 - 0.5y_3$$

$$y_2 = 3 - 5y_1 - 2z_1 + y_3$$

Step 2

(P)

$$\zeta = -7 - w_3 - 4x_2$$

$$w_1 = 18 + 2w_3 + 7x_2$$

$$x_1 = 7 + w_3 + 3x_2$$

$$w_2 = 6 + 2w_3 + 2x_2$$

(D)

$$-\zeta = 7 - 18y_1 - 7z_1 - 6y_2$$

$$y_3 = 1 - 2y_1 - z_1 - 2y_2$$

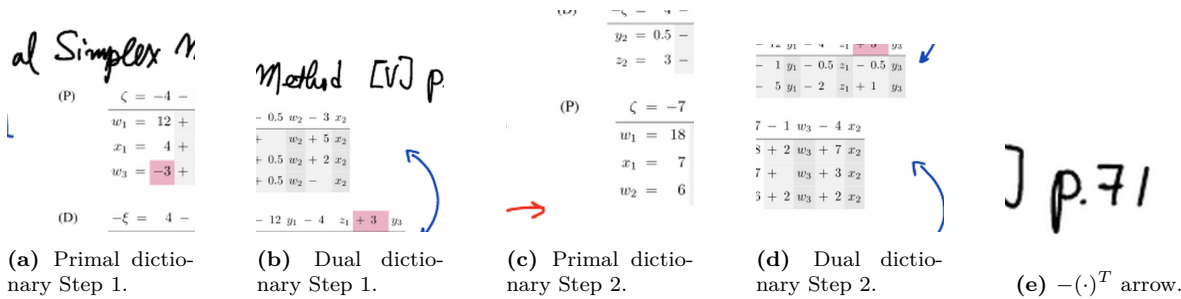
$$z_2 = 4 - 7y_1 - 3z_1 - 2y_2$$

Opt. for both

Description

Slide 4 of 15 in a Linear Programming course demonstrates two iterations of the dual simplex method. The slide shows the primal (P) and dual (D) dictionaries for two steps. Step 1 shows the initial primal dictionary with a negative right-hand side in the w_3 equation, indicating primal infeasibility. The corresponding dual dictionary is also presented. Step 2 shows the dictionaries after one iteration of the dual simplex method. The negative right-hand side in the primal dictionary has been eliminated, and the primal problem is now feasible. The dual dictionary is updated accordingly. The notation $-(\cdot)^T$ indicates that the dual dictionary is obtained by taking the negative transpose of the primal dictionary's constraint matrix. The red arrow indicates that the optimal solution is found for both the primal and dual problems after Step 2. The pink shading highlights the negative right-hand side in the primal dictionary of Step 1. The context of the course is crucial for understanding the dual simplex method and its iterative process of transforming primal and dual dictionaries to reach an optimal solution. The slide demonstrates the relationship between the primal and dual problems and how they are updated during the iterations of the dual simplex method.

Figures



Slide 5

Content

Dual Based Phase I Algorithm [V] p.73

maximize $-x_1 + 4x_2$

subject to $-2x_1 - x_2 \leq 4$

$-2x_1 + 4x_2 \leq -8$

$-x_1 + 3x_2 \leq -7$

$x_1, x_2 \geq 0$

(P)

$\zeta = -1x_1 + 4x_2$

$w_1 = 4 + 2x_1 + x_2$

$w_2 = -8 + 2x_1 - 4x_2$

$w_3 = -7 + x_1 - 3x_2$

(D)

$-\zeta = -4y_1 + 8y_2 + 7y_3$

$z_1 = 1 - 2y_1 - 2y_2 - y_3$

$z_2 = -4 + y_1 + 4y_2 + 3y_3$

origin not feasible for both

Description

Slide 5 of 15 in a Linear Programming course introduces a "Dual Based Phase I Algorithm". The slide presents a primal linear programming problem (P) in canonical form, aiming to maximize $-x_1 + 4x_2$ subject to three constraints and non-negativity conditions. Importantly, two constraints have negative right-hand sides, indicating that the origin is not primal feasible. The slide then shows the corresponding dual problem (D) in dictionary form, where the objective function is $-\zeta = -4y_1 + 8y_2 + 7y_3$. The dual dictionary also shows that the origin is not dual feasible because $z_2 = -4$. The dictionaries are presented in tabular form, clearly showing the coefficients of the variables and the constants. The red arrows and the text "origin not feasible for both" highlight the infeasibility of both the primal and dual problems at the origin, setting the stage for the application of a Phase I algorithm to find a feasible solution before proceeding to optimization.

Figures

$$\begin{array}{ll} \text{maximize} & -x_1 + 4x_2 \\ \text{subject to} & -2x_1 - x_2 \leq 4 \\ & -2x_1 + 4x_2 \leq -8 \\ & -x_1 + 3x_2 \leq -7 \\ & x_1, x_2 \geq 0. \end{array}$$

(a) (P) dictionary.

(b) (D) dictionary.

Slide 6

Content

Dual Based Phase I Algorithm [V] p.73

maximize $-x_1 + 4x_2 \xleftarrow{\text{Change obj}}$

subject to $-2x_1 - x_2 \leq 4$

$-2x_1 + 4x_2 \leq -8$

$-x_1 + 3x_2 \leq -7$

$x_1, x_2 \geq 0$

(P)

$\zeta = -1x_1 + 4x_2$

$w_1 = 4 + 2x_1 + x_2$

$w_2 = -8 + 2x_1 - 4x_2$

$w_3 = -7 + x_1 - 3x_2$

(D)

$-\zeta = -4y_1 + 8y_2 + 7y_3$

$z_1 = 1 - 2y_1 - 2y_2 - y_3$

$z_2 = -4 + y_1 + 4y_2 + 3y_3$

Now origin is D-feasible

Description

Slide 6 of 15 in a Linear Programming course continues the discussion of the Dual Based Phase I Algorithm. It presents a linear program aiming to maximize $-x_1 + 4x_2$ subject to three constraints, with two of them having negative right-hand sides, indicating primal infeasibility. A red arrow points to the objective function, indicating a change to $-x_1 - x_2$ for the Phase I algorithm. The slide then shows the primal (P) and dual (D) dictionaries. The primal dictionary shows the constraints in dictionary form, highlighting the negative right-hand sides in w_2 and w_3 , confirming primal infeasibility. The dual dictionary is also presented, showing that the origin ($y_1 = y_2 = y_3 = 0$) is dual feasible because all z_i are non-negative except for $z_2 = -4$. A handwritten note "Now origin is D-feasible" indicates that a modification has been made to the problem, resulting in dual feasibility at the origin. This likely involves changing the objective function as indicated by the red arrow and note. The context of the course is crucial for understanding the dual based Phase I algorithm and its application to solving linear programs that are infeasible in their primal form. The slide demonstrates the transition from an infeasible primal problem to a dual feasible one, setting the stage for the next steps in the algorithm.

Figures

Figure showing the primal and dual dictionaries. (a) Primal dictionary (P) shows the objective function and constraints in dictionary form. (b) Dual dictionary (D) shows the dual objective function and constraints in dictionary form. A red arrow points to the objective function in (a), indicating a change to $-x_1 - x_2$.

(a) (P) dictionary.

(b) (D) dictionary.